

Project 7

Actuation-Redundant Planar Parallel Manipulator - 6DOF

Format: Group of two students; simulation-based project.

Motivation and context

Actuation redundancy allows a robot to avoid singularities that would otherwise limit its workspace and degrade its stiffness and accuracy. Indeed, the extra actuators can be used to control internal forces and maintain a well-conditioned Jacobian even near configurations that would be singular for a non-redundant robot.

An actuation-redundant parallel robot is shown in Fig. 1. The robot is also shown in the online video <https://youtu.be/SO68ei6WjrQ>. The end-effector platform is connected to three legs by spherical joints, and each leg is able to control the position of its corresponding spherical joint. The robot is therefore actuation redundant: each leg has 3 DOFs, giving 9 actuated DOFs in total, while the end-effector platform has only 6 DOFs.

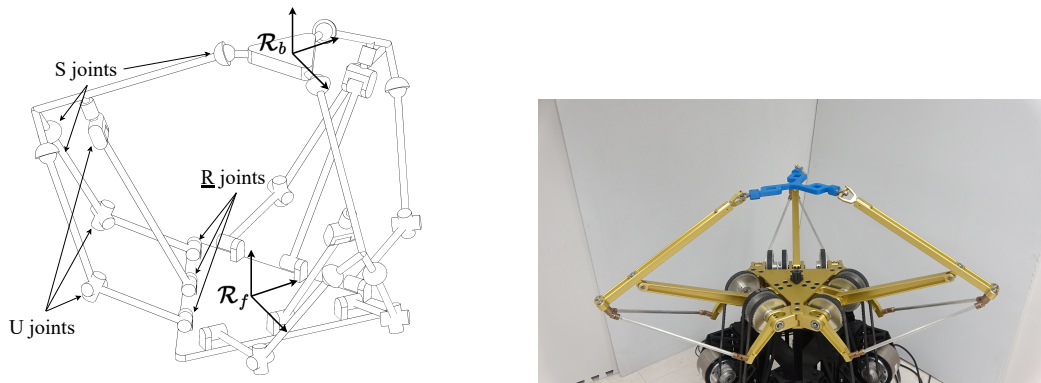


Figure 1: CAD model and prototype of the actuation-redundant robot [1].

In this project, a planar case is considered. The leg mechanisms from [1] are replaced by 5-RPR mechanisms and are connected to the end-effector by revolute joints rather than spherical joints. The resulting architecture is shown in Fig. 2.

Tasks

Overall objectives. Perform a singularity analysis of the entire robot. Demonstrate that the singularity of a single leg mechanism does not necessarily imply that the entire robot is singular.

- **Compute forward and inverse kinematics.** While inverse kinematics can be solved analytically, numerical methods can be used for forward kinematics. What is the impact of the redundant actuation on forward kinematics?

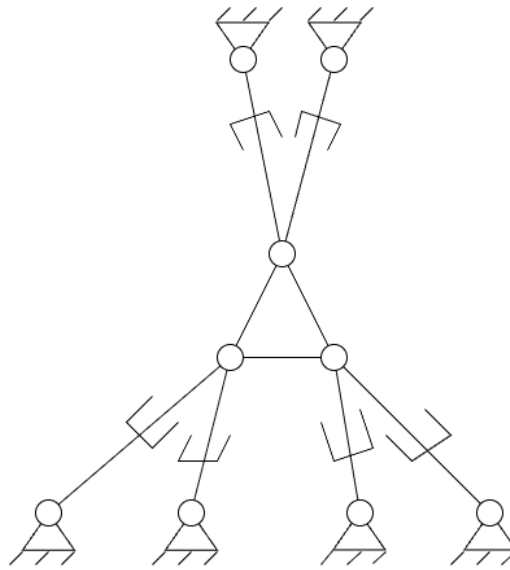


Figure 2: Kinematics architecture of the actuation-redundant robot.

- **Create a Simscape model of the actuation-redundant robot.** Use this model in order to verify your kinematics equations.
- **Analyze the singularities.** Find a trajectory (or configuration) where the robot is not singular while one 2-RPR mechanism is singular. If possible, do the same with two (respectively three) singular 2-RPR mechanisms.
- **Design a trajectory in task space.** This trajectory should illustrate the unlimited rotation capability. Is such a motion possible without mechanical interference? In other words, is unlimited rotation feasible in practice?
- **Determine joint torques.** Consider quasi-static assumption with mass associated to the end-effector and gravity. Is the solution unique? Are the results consistent with the singularity analysis?

Available documents

Paper [1] can be accessed via [Moodle](#).

Expected outcome

Students should obtain the inverse kinematics and the Jacobian matrices of the actuation-redundant robot. They should understand why the singularity of a leg does not necessarily imply that the whole robot is singular.

Submit a report, and a short video (screen recording) with Simscape illustrating one of the key results associated to the singularity analysis.

References

- [1] A. Yiğit, S. Foucault, and C. Gosselin, “Backdrivable Redundantly Actuated Parallel Robot for Sensorless Physical Human-Robot Interaction,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.